

The background features a complex network of thin grey lines and dots, forming a web-like structure. Scattered throughout are various triangles of different sizes and orientations, some with solid outlines and others with dashed or dotted outlines. The overall aesthetic is minimalist and geometric.

Research Meeting 018

Mary Ann Tan
July 23, 2021

Last Research Meeting RM017

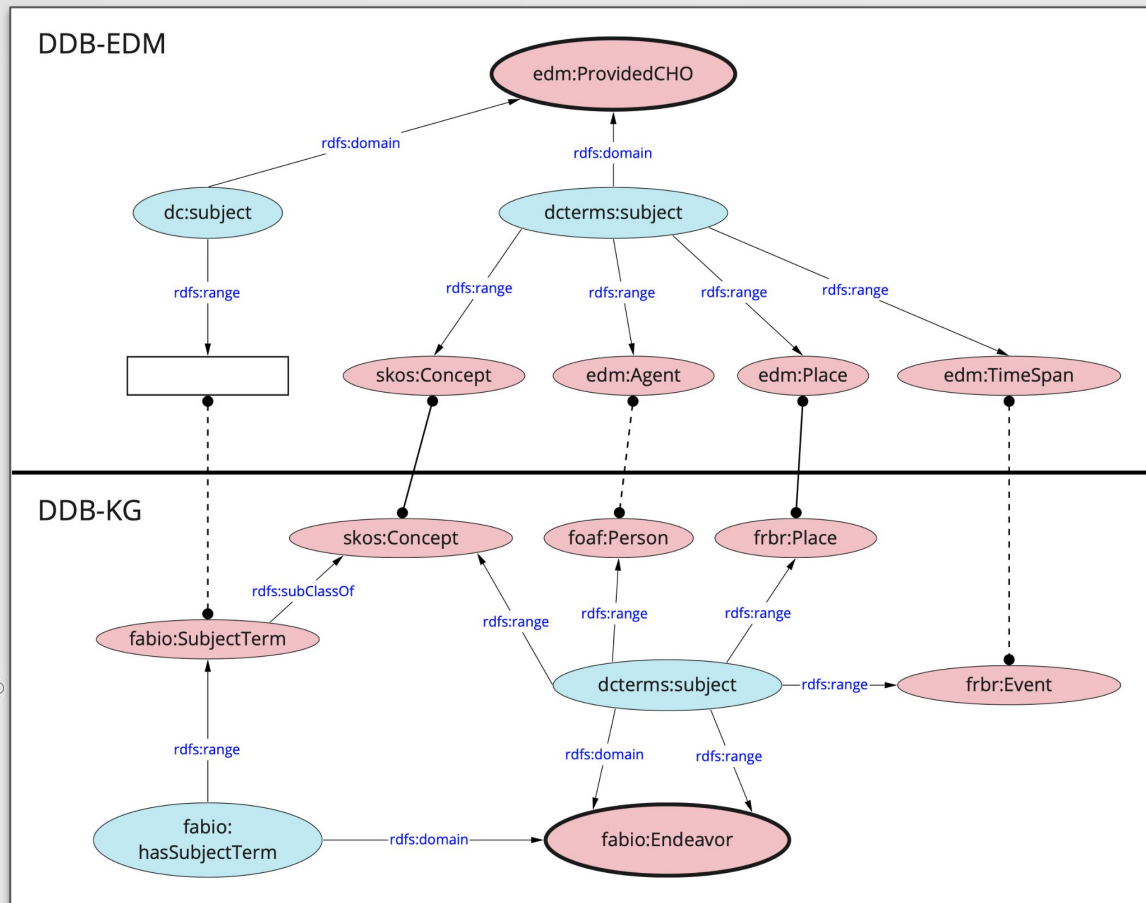
- 0. Histoinformatics Workshop Outline
 - 1. Short Recap of Subject and Subject Term
 - 2. Provenance (Prov-O)
 - 3. Ordered sequence of objects.
 - 4. Attributes of spatial and temporal ranges.
 - a. edm:TimeSpan or frbr:Event
 - b. edm:Place / frbr:Place
 - 5. Archive

Workshop Outline

1. Introduction
 - Motivation
2. Related Work
3. DDB
 - DDB-EDM
 - Challenges
4. KG Construction
 - Ontology Mapping
 - FaBiO (Libraries)
 - RiCO? (Archives)
 - Proposed Workflow
5. Future Work



Recap of Subject and Subject Term



VII. Physical and Other Attributes - 2

Attribute	DDB		FaBiO	Alternatives
	DDB-EDM	View		
Event	edm:hasMet ¹		-	
Table of Contents	dcterms:tableOfContents²	Inhaltsverzeichnis	dcterms:hasPart → doco:TableOfContents (R)	
Conformity	dcterms:conformsTo³		-	

¹ Event-centric modeling. Range = edm:Event

² Literal values or reference to other objects.

³ Range is dcterms:Standard. Not found in dataset.

VIII. ore:Aggregation Attributes

Attribute	DDB		FaBiO	Alternatives
	DDB-EDM	View		
Data Provider of DDB	edm:dataProvider	Elektronische Ausgabe	frbr:owner ¹ → foaf:Person → foaf:Organization	
Digitized Object or Webpage	edm:hasView ↳ edm:isShownBy ↳ edm:isShownAt	PURL	fabio:hasReproduction → fabio:DigitalItem	
Thumbnail	edm:object			schema:thumbnail
Data Provider of Europeana ²	edm:provider		-	
Rights to Digital Object	edm:rights		fabio:DigitalItem → dcterms:rights	
Rights to Metadata	dcterms:rights		fabio:AnalogItem → dcterms:rights	

¹ [frbr:owner](#) - An item linked to its owner.

² Value always set to DDB.

IX. edm:WebResource Attributes

Attribute	DDB		FaBiO	Alternatives
	DDB-EDM	View		
Creation Date	dcterms:created	(wann)	dcterms:created	
Creator	dc:creator		dcterms:creator	
Description	dc:description		dc:description	
Preceded By	edm.isNextInSequence ¹		-	bf:precededBy (TBD)
Part	dcterms:isPartOf		dcterms:isPartOf dcterms:hasPart	
Media Type	dc:type (audio, photo, text, video, not digitized)		fabio:isStoredOn fabio:StorageMedium (R)	dcterms:format Internet Media Type (MIME)
Rights to Digital Object	edm:rights	Rechts status	fabio:DigitalItem → dcterms:rights	
Rights to Metadata	dcterms:rights		fabio:AnalogItem → dcterms:rights	
Other Rights	dc:rights		dcterms:rights	

¹ Sequence of objects on the Item level.

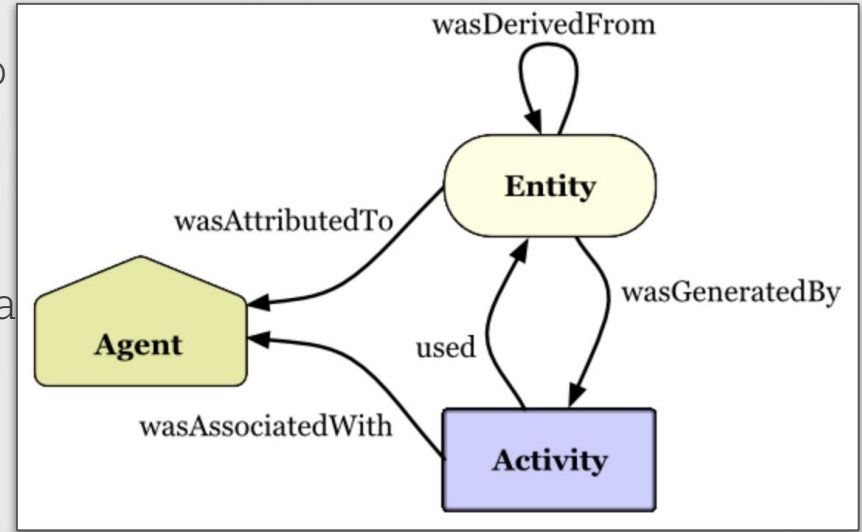
PROV

“describes how things were created or delivered”. - PROV-PRIMER

Entity - i.e. a CHO is an entity whose provenance refers to other entities (creator).

Activity - how entities come into existence and how their attributes change to become new entities. i.e. cataloging, translation.

Agent - entities that may be ascribed responsibility for an activity; can be a person, an organization or an inanimate object.
- *attributed to*: expresses an agent's responsibility for that entity.



Roles, Derivations, Time and Specializations

Roles - function of an entity in an activity; relationship between entity and activity.
- specify how agents are involved in an activity.
- application-specific.

Derivation - when an entity's existence, content, characteristics are partly due to another entity. i.e. revision, quotation.

Specialization - linking different descriptions of the same thing.
- i.e. linking descriptions (alternates) of the same CHO from different providers.



Provenance in DDB

DDB Output XML File elements:

```
<ns4:cortex xmlns:ns4="http://www.deutsche-digital  
  <ns4:properties>...</ns4:properties>  
  <ns4:edm>...</ns4:edm>  
  <ns4:indexing-profile>...</ns4:indexing-profile>  
  <ns4:preview media="text" type="Kultur">...</ns4:preview>  
  <ns4:view>...</ns4:view>  
  <source:source xmlns:source="http://www.deutsche-digital  
  <ns4:provider-info>...</ns4:provider-info>  
  <ns4:aggregator-info>...</ns4:aggregator-info>  
  <ns4:binaries>...</ns4:binaries>  
</ns4:cortex>
```

```
"Prov": {  
  "item-id": "JZ2PWIS62TWZJN5F75QX276VEMHIQCNH",  
  "dataset-id": "26211197755718865YlhD",  
  "dataset-label": "Gesamtlieferung - SLUB Dresden (00014072) - METS/MODS",  
  "revision-id": "6",  
  "ingest-date": "2021-04-21T07:09:54+0200",  
  "cortex-type": "Kultur",  
  "mapping-version": "1.163",  
  "source-ref": "LVWQHfHZFAG3YPYISJ6HBOF7K3YYA2MQ",  
  "source-type": "http://www.loc.gov/METS/",  
  "provider": {  
    "name": "Sächsische Landesbibliothek - Staats- und Universitätsbibliothek  
    "uri": "http://www.slub-dresden.de",  
    "id": "00014072",  
    "ddbid": "265BI7NE7QBS4NQMZCCGIVLFR730C0SL",  
    "isil": "http://ld.zdb-services.de/resource/organisations/DE-14"  
  }  
}
```

PROV-O

Activity

Agent

, and

Entities

```
"Prov": {  
  "item-id": "JZ2PWIS62TWZJN5F75QX276VEMHIQCNH",  
  "dataset-id": "26211197755718865YlhD",  
  "dataset-label": "Gesamtlieferung - SLUB Dresden (00014072) - METS/MODS",  
  "revision-id": "6",  
  "ingest-date": "2021-04-21T07:09:54+0200",  
  "cortex-type": "Kultur",  
  "mapping-version": "1.163",  
  "source-ref": "LVWQHfHZFAG3YPYISJ6HBOF7K3YYA2MQ",  
  "source-type": "http://www.loc.gov/METS/",  
  "provider": {  
    "name": "Sächsische Landesbibliothek - Staats- und Universitätsbiblio",  
    "uri": "http://www.slub-dresden.de",  
    "id": "00014072",  
    "ddbid": "265BI7NE7QBS4NQMZCCGIVLFR730COSL",  
    "isil": "http://ld.zdb-services.de/resource/organisations/DE-14"  
  }  
}
```

Ingest

XSLT

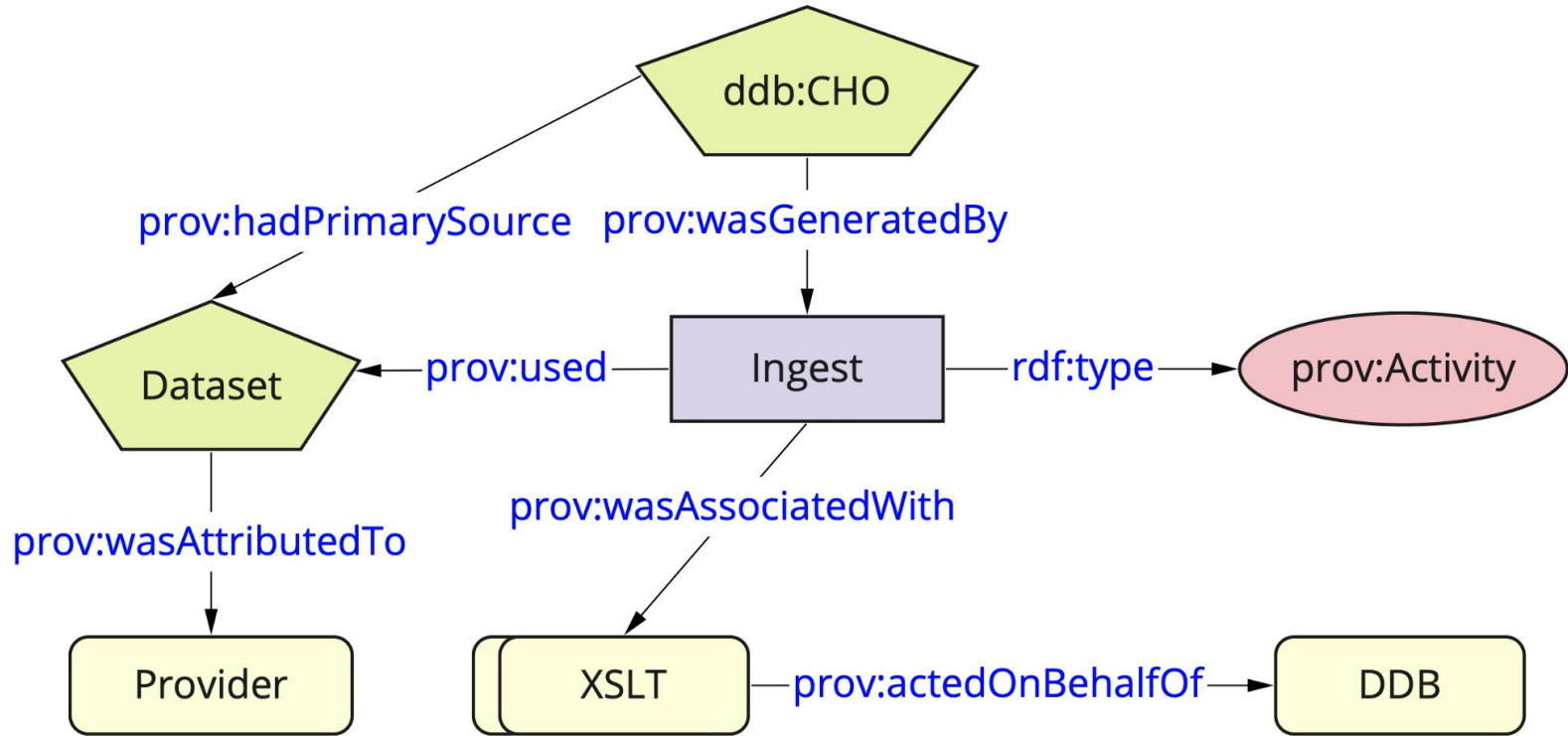
DDB

Provider

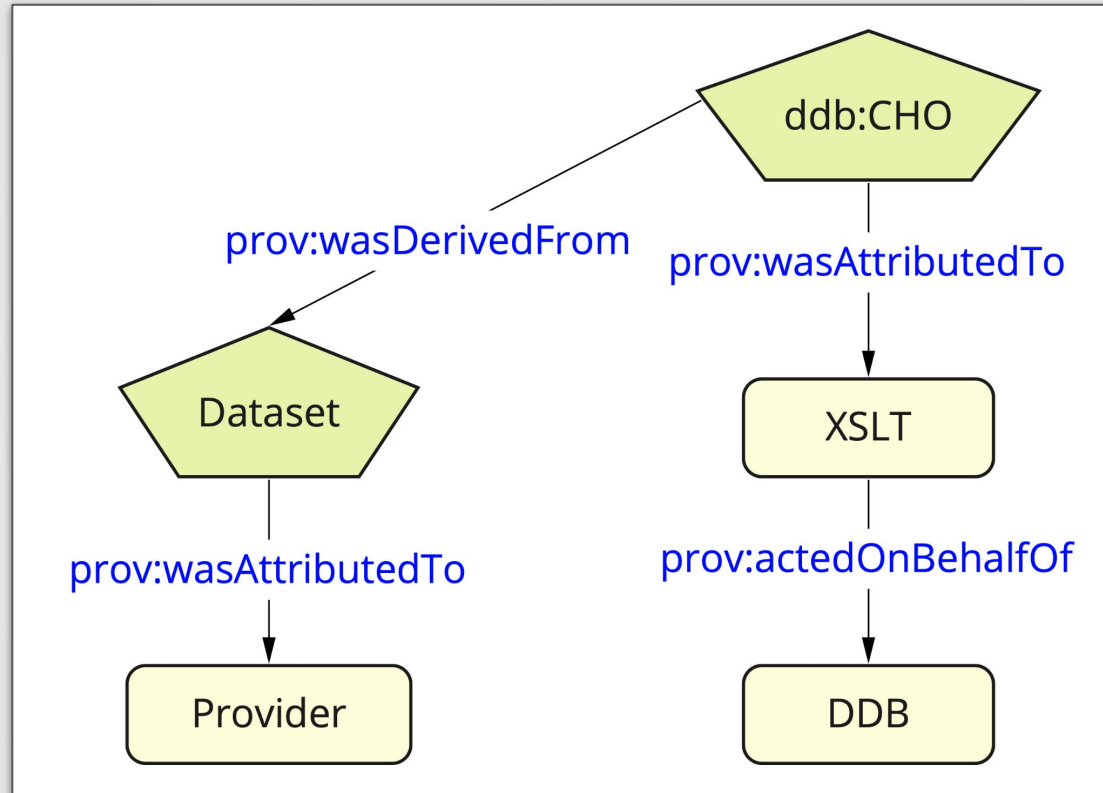
CHO

Dataset

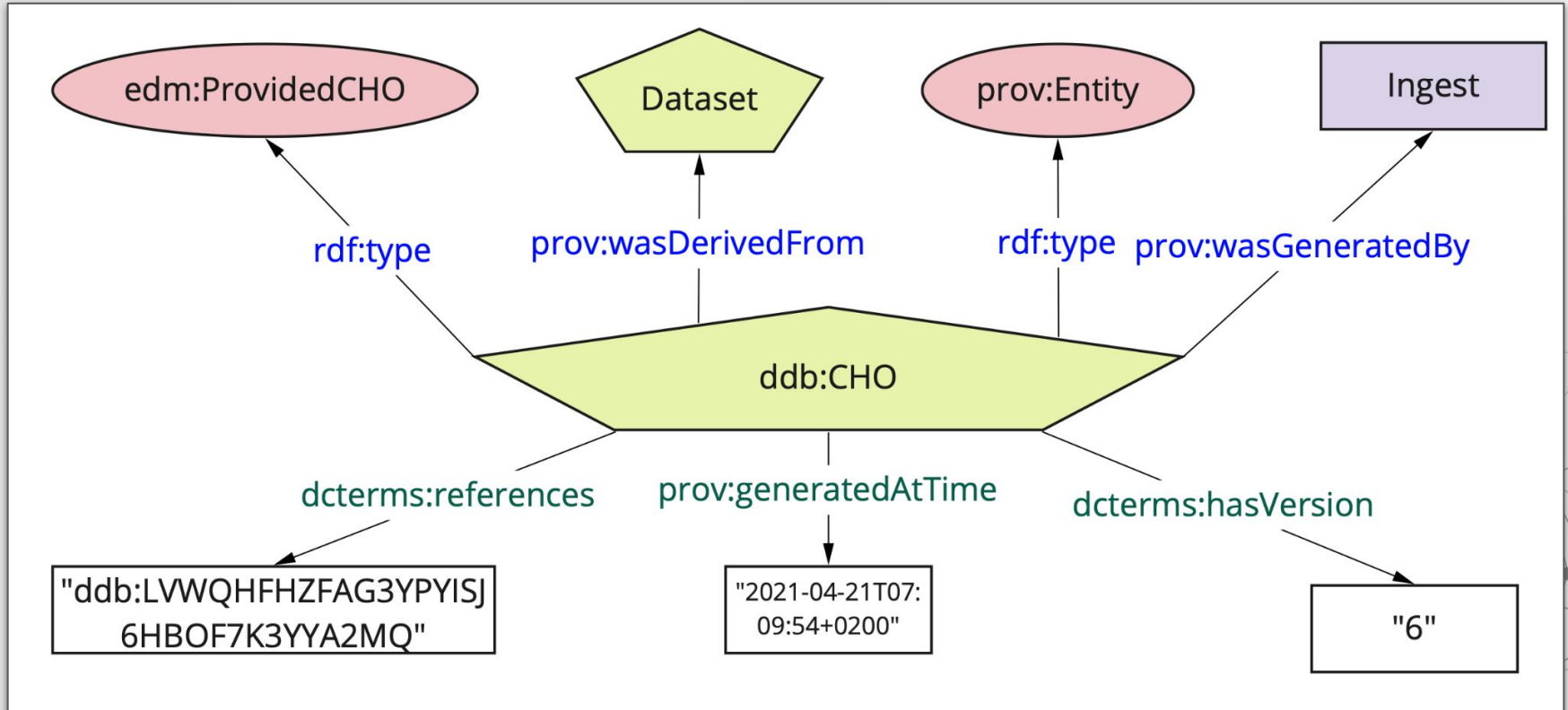
Ingest Activity



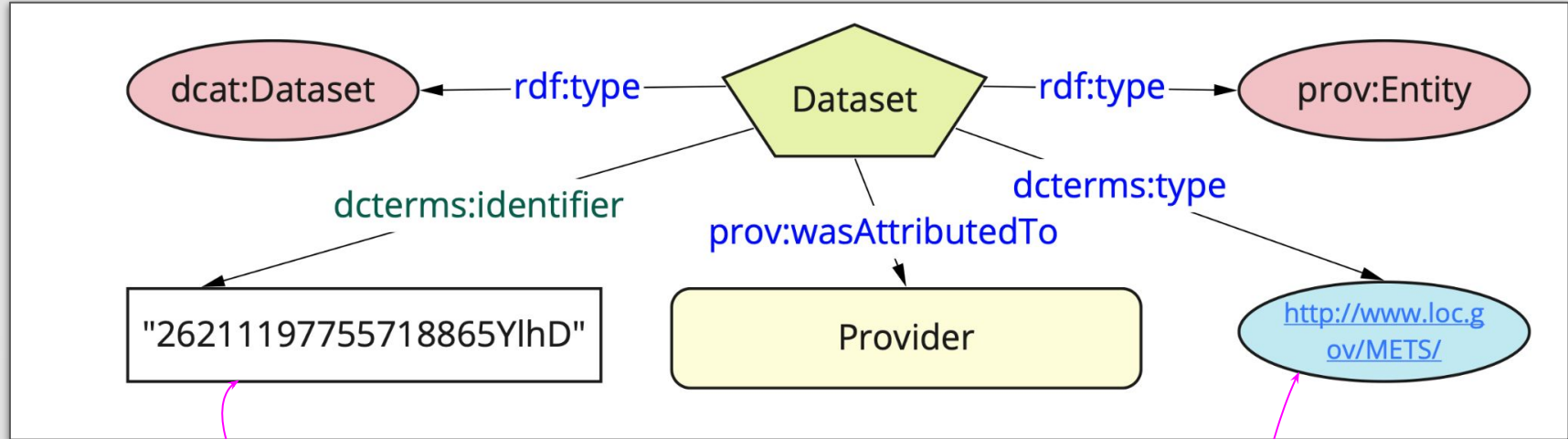
Without Activity



Entities: CHO

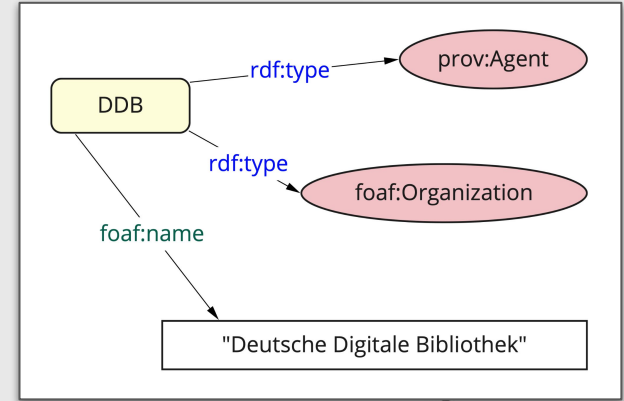
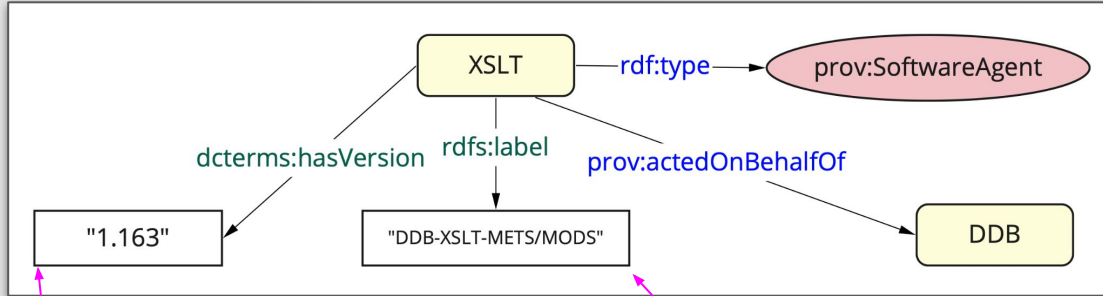


Entities: Dataset



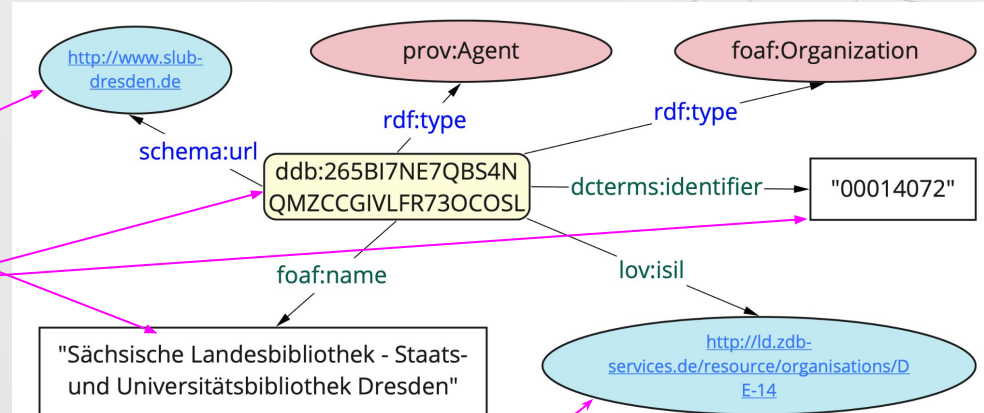
```
"Prov": {  
  "item-id": "JZ2PWIS62TWZJN5F75QX276VEMHIQCNH",  
  "dataset-id": "26211197755718865YlhD",  
  "dataset-label": "Gesamtlieferung - SLUB Dresden (00014072) - METS/MODS",  
  "revision-id": "6",  
  "ingest-date": "2021-04-21T07:09:54+0200",  
  "cortex-type": "Kultur",  
  "mapping-version": "1.163",  
  "source-ref": "LVWQHfHZFAG3YPYISJ6HB0F7K3YYA2MQ",  
  "source-type": "http://www.loc.gov/METS/",
```


Agents



```

"Prov": {
  "item-id": "JZ2PWIS62TWZJN5F75QX276VEMHIQCNH",
  "dataset-id": "26211197755718865Ylhd",
  "dataset-label": "Gesamtlieferung - SLUB Dresden (00014072) - METS/MODS",
  "revision-id": "6",
  "ingest-date": "2021-04-21T07:09:54+0200",
  "cortex-type": "Kultur",
  "mapping-version": "1.163",
  "source-ref": "LVWQHfHZFAG3YPYISJ6HBOF7K3YYA2MQ",
  "source-type": "http://www.loc.gov/METS/",
  "provider": {
    "name": "Sächsische Landesbibliothek - Staats- und Universitätsbibliothek Dresden",
    "uri": "http://www.slub-dresden.de",
    "id": "00014072",
    "ddbid": "265BI7NE7QBS4NQMZCCGIVLFR73OCOSL",
    "isil": "http://ld.zdb-services.de/resource/organisations/DE-14"
  }
}
  
```





02

Ontology for Museum

Rijksmuseum (Dijkshoorn et al., 2018a, b)

- From VRA Core → EDM (Dublin Core + SKOS)
 - `dc:type - skos:Concept` linked to Getty's AAT, own thesaurus (rma), literals
 - `dc:subject - skos:Concept` to AAT, Iconclass (SKOS)
- Authors summarized **challenges** using EDM and CIDOC-CRM through of the following example objects:

Maerten Soolmans



Source: [Christie's](#)

Oopjen Coppit



Source: [Rijksmuseum](#)

Challenges: Visual Arts Objects

Maerten Soolmans



Oopjen Coppit



Source: [Christie's](#)

Rembrandt's Wedding Portraits:

- previously privately-owned (Rothschild)
- sold to Rijksmuseum and Louvre separately
- always exhibited together, moved every 5 years
- multiple fitting frames
- Coppit married twice: *Maarten Soolmans* and *Maerten Daey*

1. Change of Ownership
2. Change of Location
3. Change of Related Objects
4. Conflicting Information (Rijksmuseum's Marten vs Louvre's Maerten)

Challenges: Historical Objects



Source: [Rijksmuseum](https://www.rijksmuseum.nl/en/collections/object/1111111111)

Napoleon's Pistols with case and accessories

- pistols created by Jean Le Page around 1808
- found encased in the case with accessories after the Battle of Waterloo
- case and accessories created by different people at different points in time
- one pistol depicts an eagle, adorned with engravings

1. Different types of objects.

2. Multiple contributors (agents).

3. Multiple representations (of pistols, accessories, case).

4. Representation of subject matter (of the pistol, of the depiction on the pistol).

Challenges - 1 (Dijkshoorn et al., 2018b)

Challenges	Description	Solution
Specialization	Collection-specific modeling - accuracy vs interoperability	1. Use of subclasses (cf. FaBiO) 2. Use of subproperties - wasMadeFor > wasPainterFor 3. Typing (dc:type): - controlled vocab - SKOS
Object vs Event-centric	efficiency vs completeness Object-centric - context connected to the object (agent, spatial, temporal) - <i>one-point-in-time</i> description Event-centric - composition of [events] - longer paths to connect context to object	Combination as in EDM:

Challenges - 2 (Dijkshoorn et al., 2018b)

Challenges	Description	Solution
Temporality	Changes over time: 1. Change of ownership 2. Change of location 3. Additions to or removal from the original object.	1. Subproperties - <i>currentOwner</i>, <i>previousOwner</i> 2. List of events: - perhaps, implemented as RDF collections and containers
Multiple representations and descriptions	An object may be digitized as: - image, video, 3D model, etc. Each representation may differ: - file encoding, resolution, equipment	<pre> graph TD A([ore:Aggregation]) -- "1:n edm:isShownAt edm:isShownBy edm:hasView" --> B([edm:WebResource]) A -- "1:1 edm:aggregatedCHO" --> C([edm:ProvidedCHO]) B -- "tba" --> D[TBA] B -- "tba" --> E((TBA)) B -- "dc:type" --> F[Media Type] </pre> The diagram illustrates an RDF graph structure. At the top is a pink oval labeled 'ore:Aggregation'. It has two outgoing arrows: one to a pink oval labeled 'edm:WebResource' with the property 'edm:isShownAt' (value '1:n') and another to a pink oval labeled 'edm:ProvidedCHO' with the property 'edm:aggregatedCHO' (value '1:1'). The 'edm:WebResource' node has three outgoing arrows: one to a light blue rectangle labeled 'TBA' with the property 'tba', one to a pink oval labeled 'TBA' with the property 'tba', and one to a light blue rectangle labeled 'Media Type' with the property 'dc:type'.

Challenges - 3 (Dijkshoorn et al., 2018b)

Challenges	Description	Solutions
Multiple sources of metadata that may be conflicting	Metadata relating to the same object may come from different institutions.	EDM uses proxy (not used in DDB) <pre> graph LR subgraph ViewBox [View] Proxy([Proxy]) end subgraph RepresentationBox [Representation] Rep([Representation]) end subgraph AggregationBox [Aggregation] Agg([Aggregation]) end subgraph ArtifactBox [Artifact] ProvidedCHO([Provided CHO]) AggregatedCHO([aggregated CHO]) end Proxy -.-> proxy for ArtifactBox Rep -.-> proxy in Agg Rep -.-> has view, is shown at Agg Agg -.-> aggregated CHO ArtifactBox </pre> Source: Figure 4 (Dijkshoorn et al., 2018b)
Contextualization using subject matter	Clear distinction between perceptual properties: ● materials ● dimensions AND conceptual properties (depiction): ● who ● what ● where	1. Range definition for subject 2. Use controlled iconography descriptions (<u>Iconclass</u>) 3. Cover multilayer descriptions: “Monet in his Studio Boat” - by Edouard Manet - depiction of Monet and his painting “<u>The Bridge at Argenteuil</u>”

03

ArCo



Development of ArCo

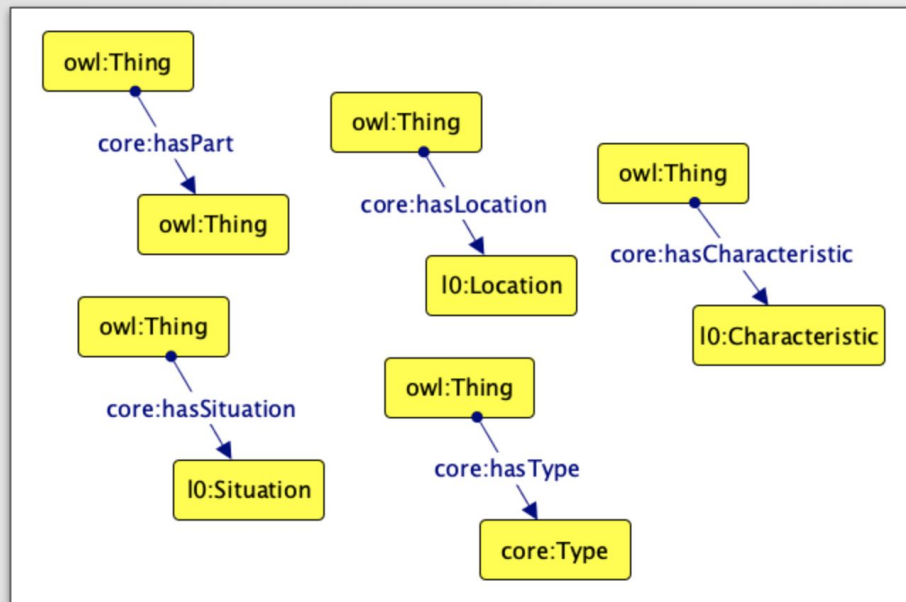
- Followed eXtreme Design (XD)
- Composed of Design + Testing + Customer teams
- Design + Customer composed of **experts** from the ICCD (Italian Central Institute for Cataloging and Documentation).
- Data is compliant with ICCD Cataloging Standards
- Main purpose is for cataloging of Italian CH, *excluding* libraries and archives.
- Uses Top-Level Ontologies Cultural-ON and OntoPiA
 - ontology network for Italian Public Administrations

Cultural Entity



- Network of ontologies meant to represent **cultural entities** and their **cataloging records**
- **arco:CulturalProperty** ~ edm:ProvidedCHO
- **arco:HistoricOrArtisticProperty** for Visual Art Objects (VAO) and other tangible/movable objects

ArCo: Core Modules

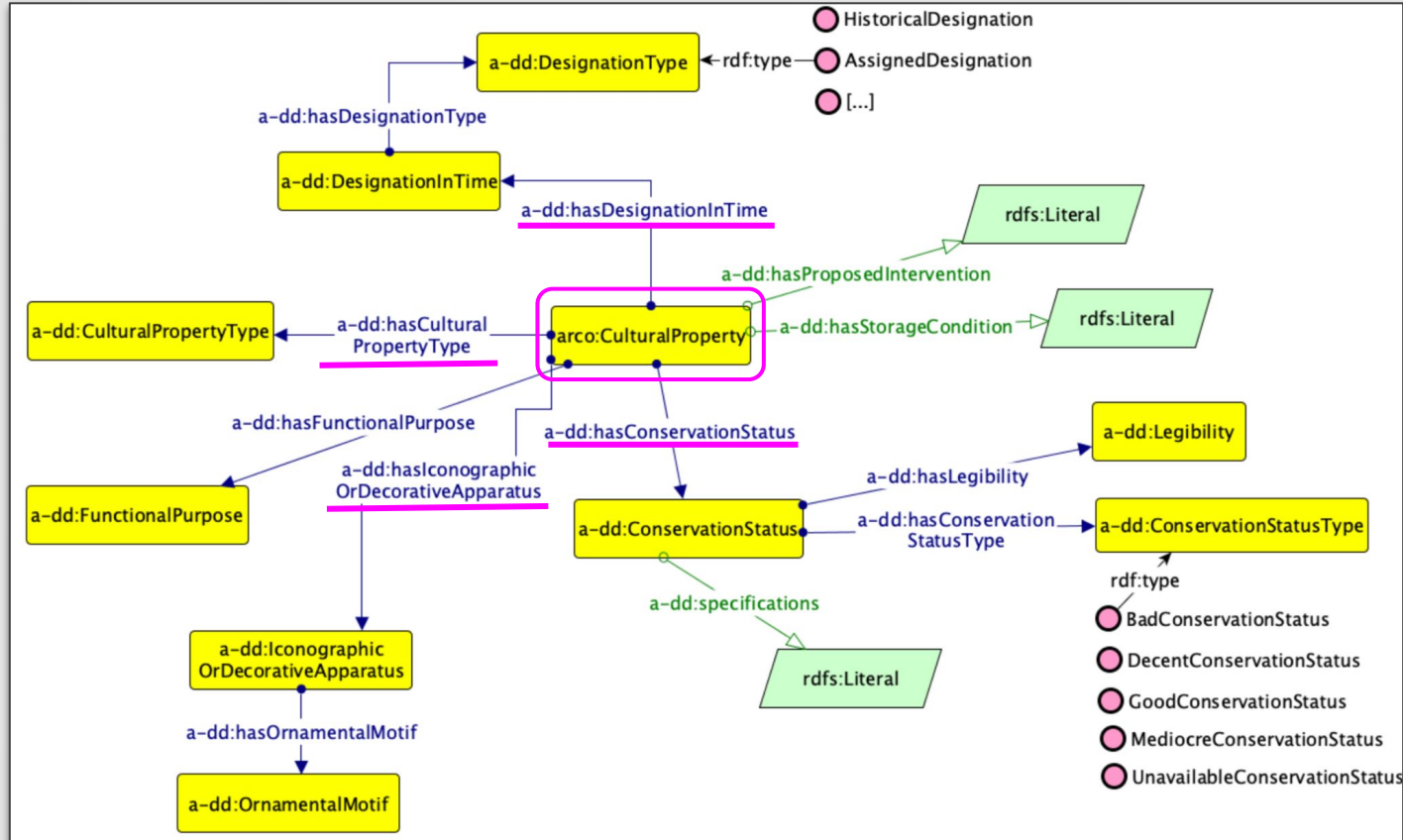


Core patterns:

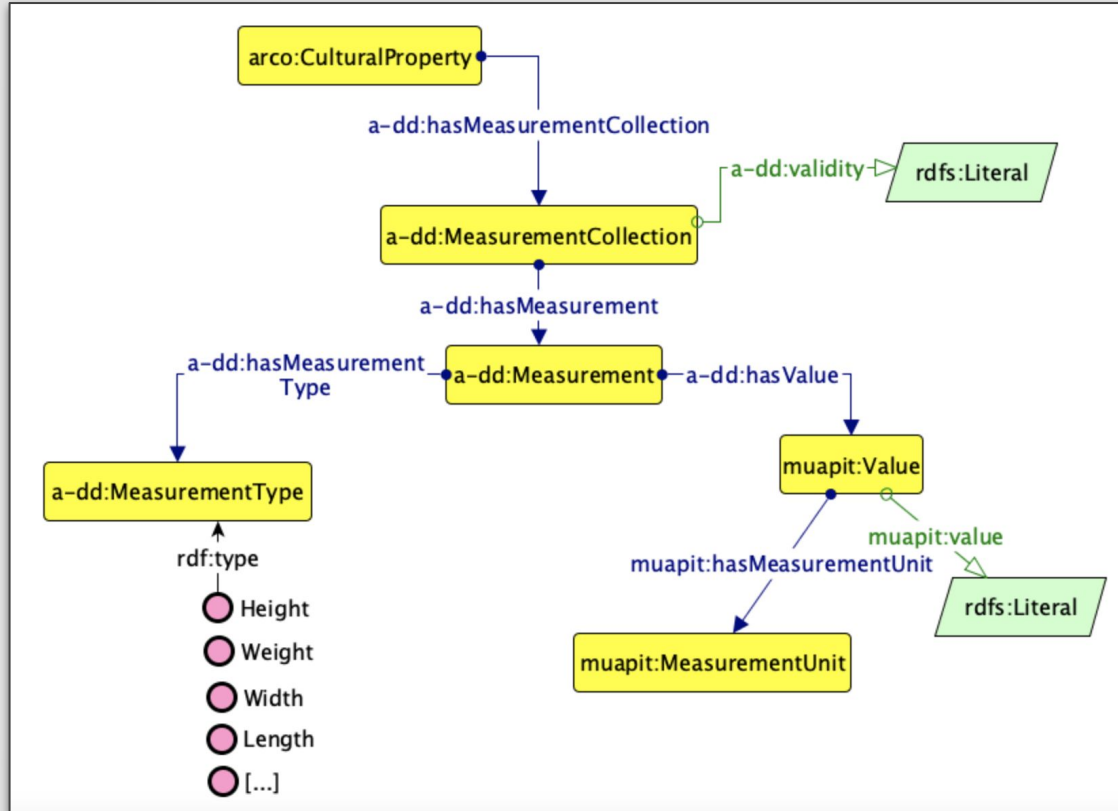
- composition (hasPart)
- situation (for contextualization)
- location
- classification (hasType)
- characteristics



Denotative Descriptions of Cultural Property



Measurements



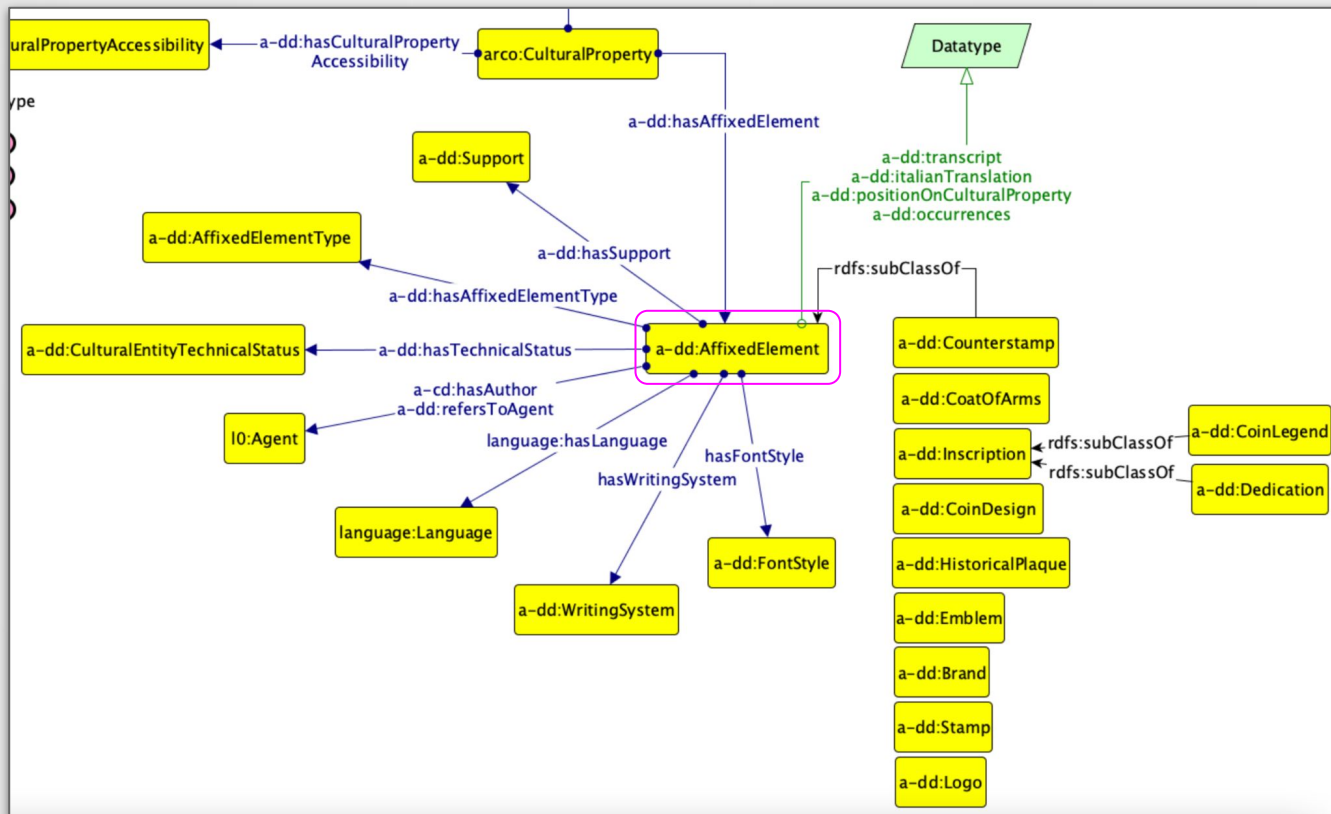
More **extensive** compared to VRA's object properties defined for measurements.

Concept of Type in ArCo

- 55 Types

- Type - Acquisition type - Additional form type - Affixed element type - Age type - Alteration type - Alternative dating type - Anomaly type - Bibliography type - Cadaastre type - Campaign type - Change of availability type - Chondrule type - Circumstance type - Communication medium type - Conservation Status Type - Contact type - Covering type - Cultural Institute or Site type - Cultural property Type	- Derivated work type - Designation Type - Documentation type - Eruptive type - Export import certification type - Finding context type - Foundation type - Horizontal element type - Iconographic Or Decorative Apparatus Type - Intervention type - Location Type - Measurement Type - Mineral deposit type - Observation survey type - Old town administrative type - Organization type - Outcrop type - Photographic heritage classification type - Preparation type - Preservation type	- Responsibility type - Roof type - Sample collected type - Site Type - Substrate Type - Survey image type - Survey type - Texture type - Title Type - Type of type specimen - Use type - Use type specification - Vertical connection type - Vertical element type - Water system type - Window type
---	--	---

Inscription and Additions



Takeaway

- ArCo is extensive but has a different purpose than DDB (cataloging vs publishing).
- Class and property names are very technical.

Challenges (Dijkshoorn et al., 2018b)	ArCo Implementation
Specialization	✓ Extensive (subclasses of Types)
Object-/Event-centric	✓ Object-centric + Event-centric (TimeIndexed*)
Temporarity	✓ DesignationInTime, TimeIndexed* (Event, Role, Classification), CatalogRecordVersion
Multiple Representations	TBD
Multiple Metadata Sources	TBD
Subject Matter	✓ IconographicOrDecorativeApparatus, OrnamentalMotif, hasSubject ⁻¹ , hasLocationAsSubject ⁻¹

Challenges in the DDB....

- Challenges described affect research and exploration use-case.

Challenges (Dijkshoorn et al., 2018b)	Applicability in the DDB
Specialization	✓ all objects are of the same class, agents with different roles are all grouped together as <code>dc:contributor</code>
Object-/Event-centric	✓ Object-centric + Event-centric (TimeIndexed*)
Temporarity	Information such as those being modeled by the Rijksmuseum are not available in the output XML file
Multiple Representations	✗
Multiple Metadata Sources	✓
Subject Matter	✓

REFERENCES

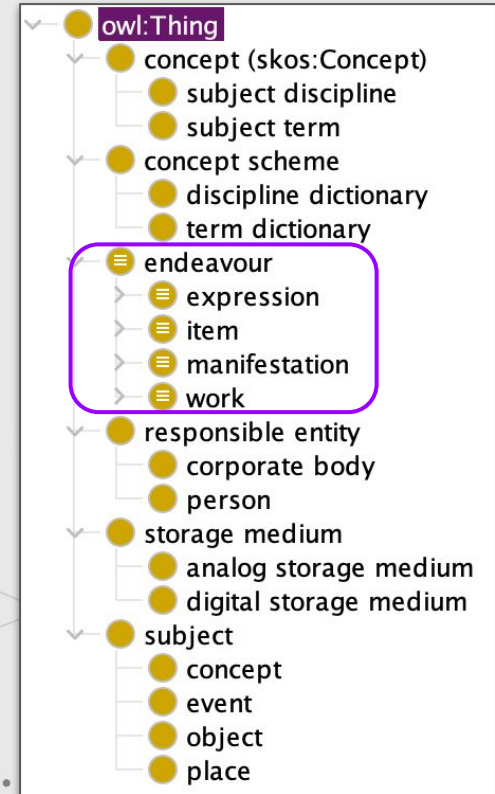
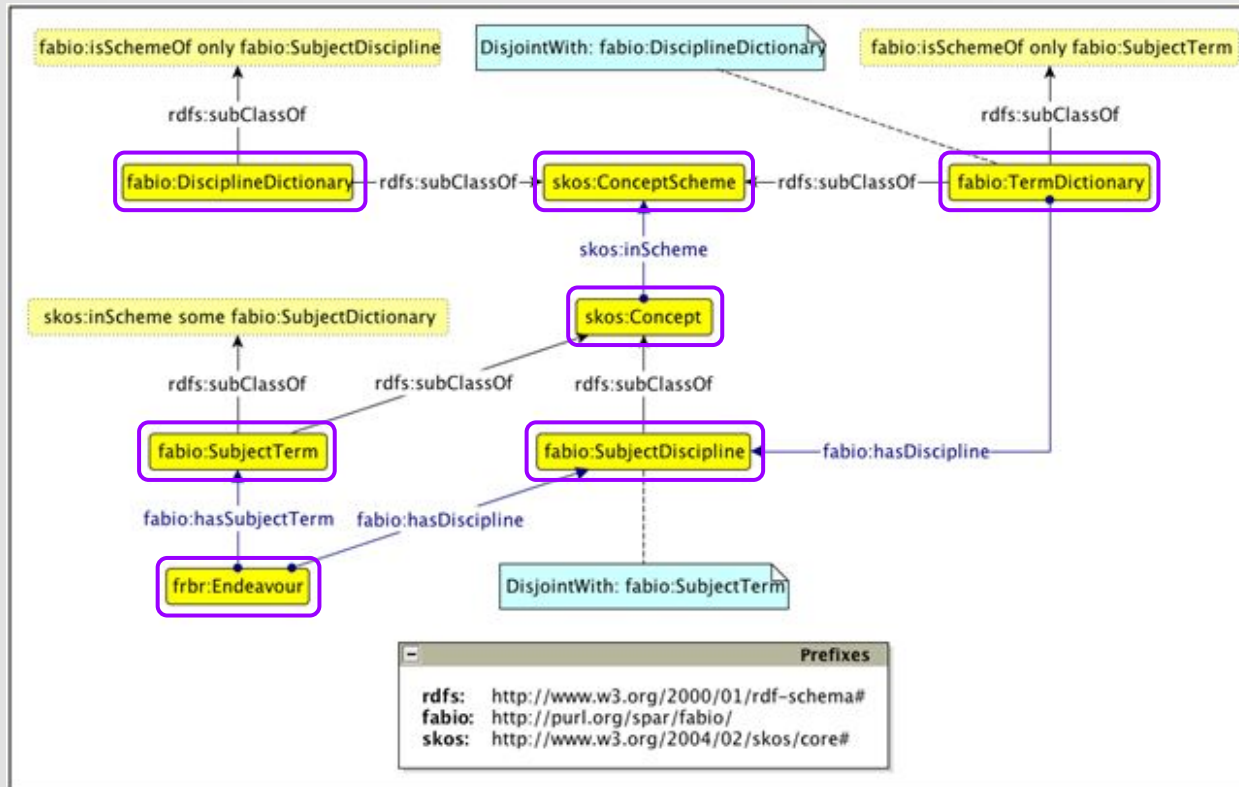
- [1] Angjeli et al. (2012). *"D5.1 Report on the alignment of library metadata with the European Data Model."*
- [2] Biagetti, Maria Teresa. (2017). *"A Comparative analysis and evaluation of bibliographic ontologies."*
- [3] Dijkshoorn, C., Arroyo, Longma, L., Lora., van Ossenbruggen, J., Schreiber, Guus, ter Weele, W., and Wielemaker, J. (2018a). *"The Rijksmuseum Collection as Linked Data."*
- [4] Dijkshoorn, C., Arroyo, Lora., van Ossenbruggen, J., and Schreiber, Guus. (2018b). *"Modeling cultural heritage data for online publication."*
- [5] Peroni, S., and Shotton, D. (2012). *"FaBiO and CiTO: Ontologies for describing bibliographic resources and citations."*
- [6] Shotton, David. (2011). *"Categorising bibliographic resources with FaBiO and SKOS."* Last Accessed: March 25, 2021.

99

EXTRAS

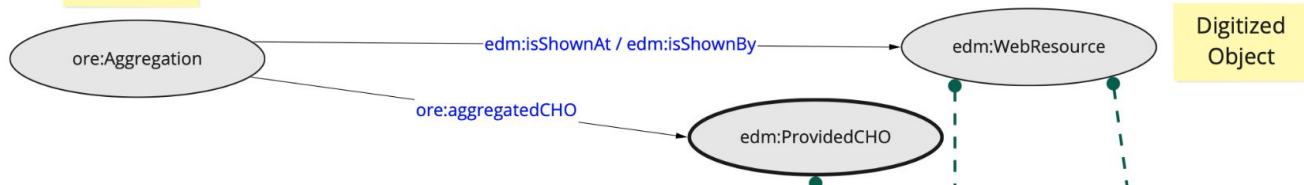


FaBiO and SKOS Concepts

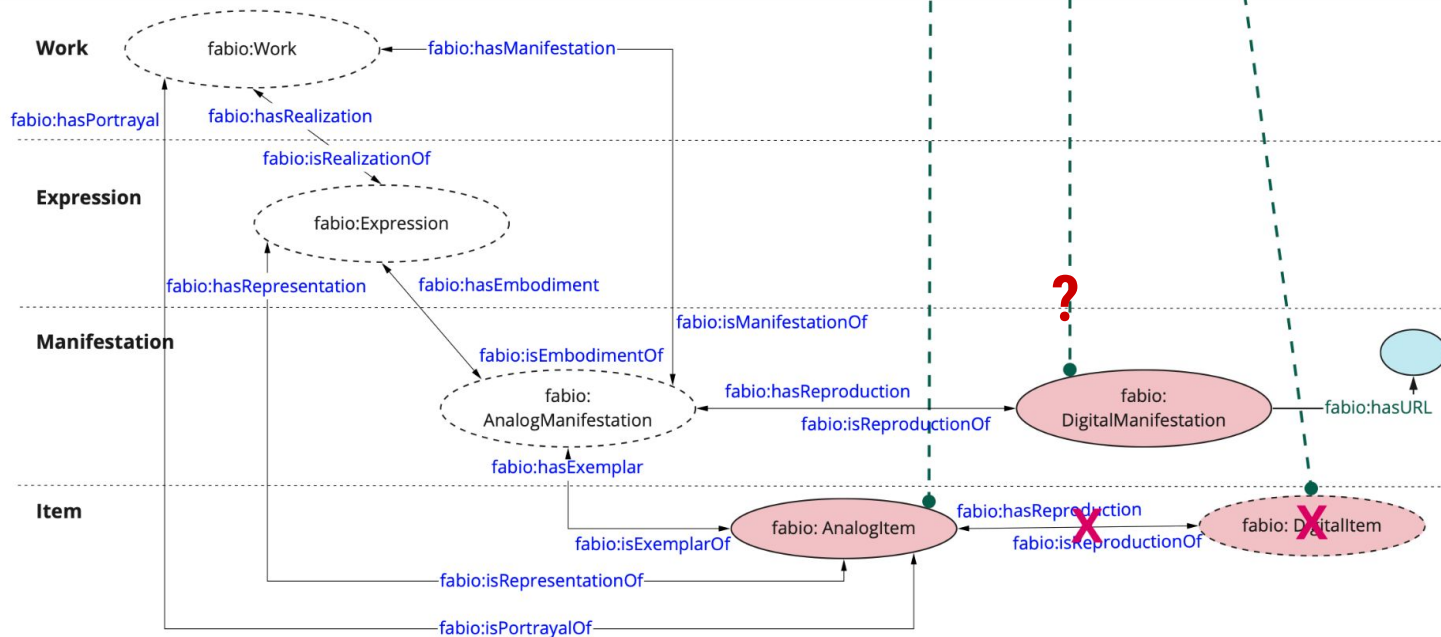


DDB-EDM

Associates the CHOs with its Digitized Manifestation



FaBiO



Challenges - 1 (Dijkshoorn et al., 2018b)

1. Specialization

- tradeoff between interoperability
- Approaches:
 - a. use of subclasses (cf. FaBiO)
 - b. *typing* using controlled terminologies + SKOS
 - c. use of subproperties - *wasMadeFor* > *wasPainterFor*

2. Object- AND Event-Centric modeling

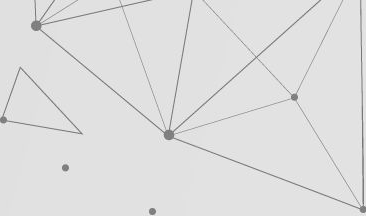
- object-centric:
 - context is directly connected to the object (agent, temporal, spatial)
 - describes an object *at one point in time* (i.e. has current owner, has current location, etc).
- event-centric:
 - increased granularity through composition [of events]
 - longer paths to connect context to the object

Challenges - 2 (Dijkshoorn et al., 2018b)

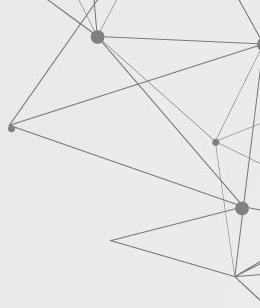
3. Temporality - capture changes over time
 - relevant for part additions (cf. Napoleon's pistol) and provenance
 - i.e. list of owners and locations; often enumerated in `dc:description`
 - Approaches:
 - a. ownership - `rdf:first`, `rdf:rest`
 - b. specific properties - `hasCurrentOwner`, `hasFormerOwner`
4. Separate description of CHO and representations
 - representations such as images, sounds, videos and 3D models
 - `(ore:Aggregation + edm:isShownAt, edm:isShownBy, edm:hasView)`
 - similarity between objects

Challenges - 3 (Dijkshoorn et al., 2018b)

5. Multiple Metadata and Views
 - from different sources that may have conflicts
6. Subject Matter
 - perceptual properties: materials, dimensions
 - conceptual properties: who, what, where is depicted



Alignment of EDM to FRBR (Angjeli et al., 2012)



- data and object properties alignment to FRBR endeavors